1 B.Tech
CSE Core
Section - A

ASSIGNMENT-1

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[DBMS UNIT-1]

Q1. What is DBMS? Why is it required?

Ans. DBMS is a software that stores, manages & retrieves data efficiently. It is required for data security, consistency, reduced redundancy and easy access.

Q2. Why is DBMS better that the traditional file system?

Ans. DBMS reduces redundancy, avoids inconsistency, provides security, support concurrency, backup & recovery.

Q3. What is data abstraction? Explain its levels.

Ans. Data abstraction means hiding the internal details of the database & showing only the necessary information to users.

There are 3 levels:-

① Physical level
→ lowest level
→ Describes how data is stored in database
eg. how records are stored on disk.

② Logical Level

→ Middle level

→ Describes what data is stored & relationship between data.

eg. Tables, attributes & relationships.

③ View Level

→ Highest level

→ Shows only required data to users.

→ Different users can see different view of same database.

Q4. What is three-schema architecture? Why is it important?

Ans. Three-schema architecture is a database design model that divides the database into 3-levels to separate user view ~~fo~~ from physical storage.

It includes:-

① Internal level → describes how data is physically stored in database.

② Conceptual level → describes the overall structure of the database.

③ External level → describes how users view the data.

It is important because it :-

→ provides data independence

→ Hides storage details from users

→ Improves security

→ Makes database design more structured & organized

Q5. What is data independence? Why is it needed?

Ans. Data independence is the ability to change the database structure at one level without affecting other levels.

Types :-

① Physical data independence → Ability to change the physical storage without affecting the logical structure.

② Logical data independence → Ability to change the logical structure without affecting user views.

It's need :-

→ Makes system flexible

→ Allows easy modification

→ Reduces application changes

→ Improves maintenance

Q6. What is an ER model?

Ans. The ER model is a conceptual data model that represents.

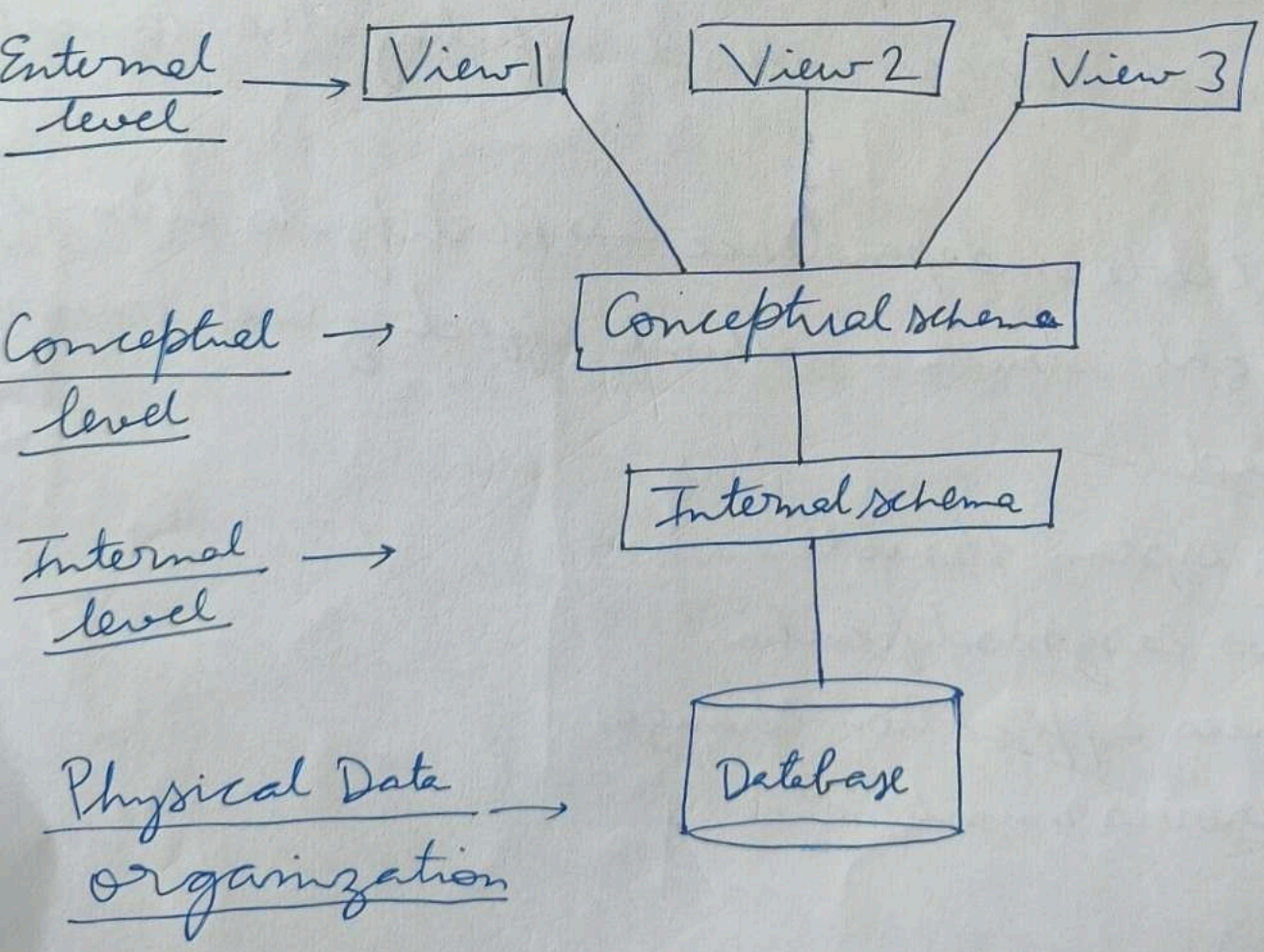
1) Entities (real world objects).
2) Their attributes
3) The relationship between them using diagrams.

It is used for Database Design.

SECTION-B

Q1. Draw the 3-schema architecture & explain each level.

Ans.



1) External level → shows different user views.

2) Conceptual level → Defines tables, attributes & relationships.

3) Internal level → Describes how data is stored in memory.

Q2. ER diagram for library management system.

Ans. 1) Entities → 1) Book (Book ID, title, Author, Publisher).

2) Member (Member ID, Name, Address).

3) Issue (Issue Date, Return Date)

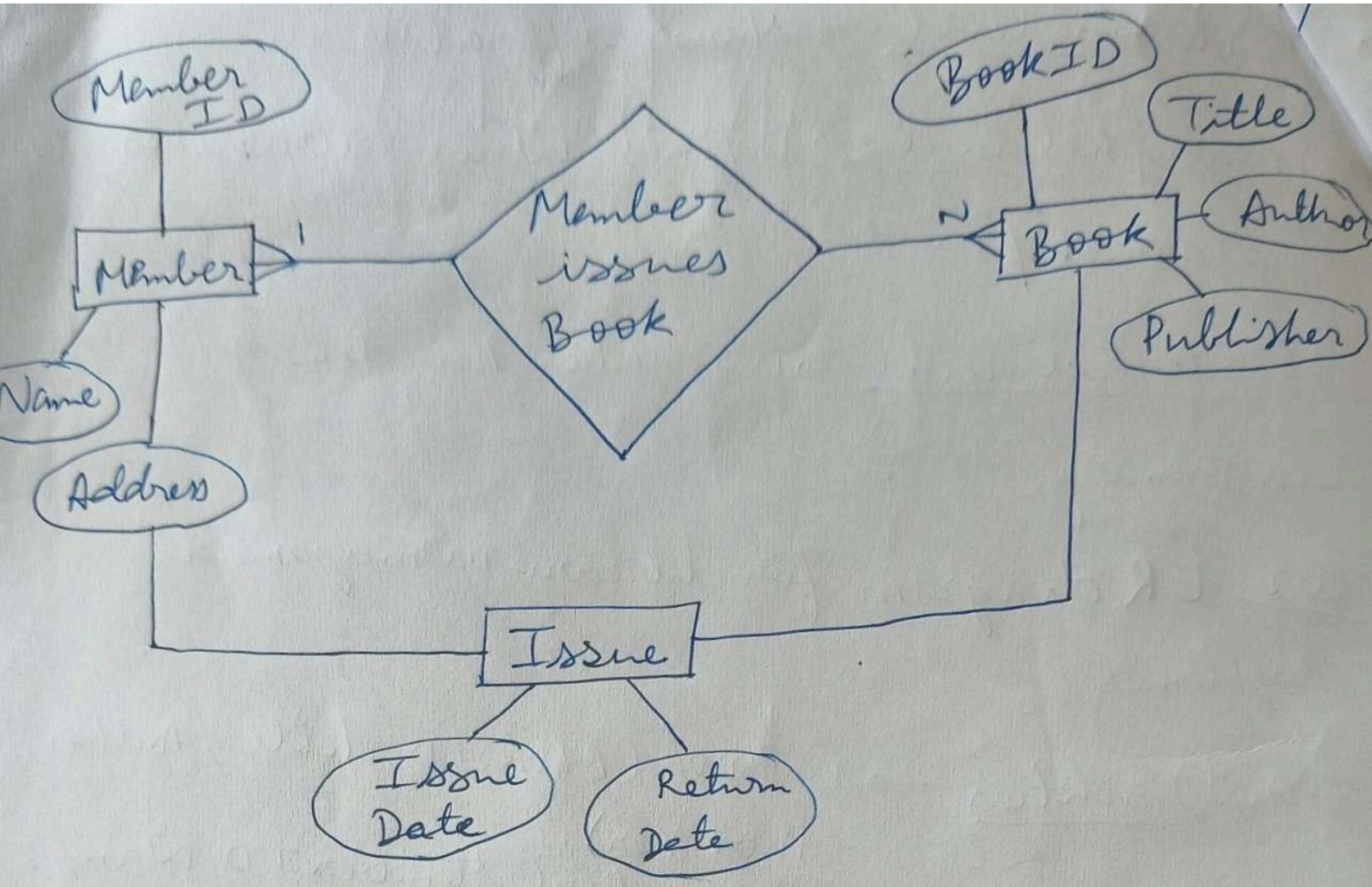
2) Relationship → Member issue book

3) Cardinality → 1) One member can issue many books.

2) One book can be issue to one member at a time.

→ Diagram :-

(PTO)

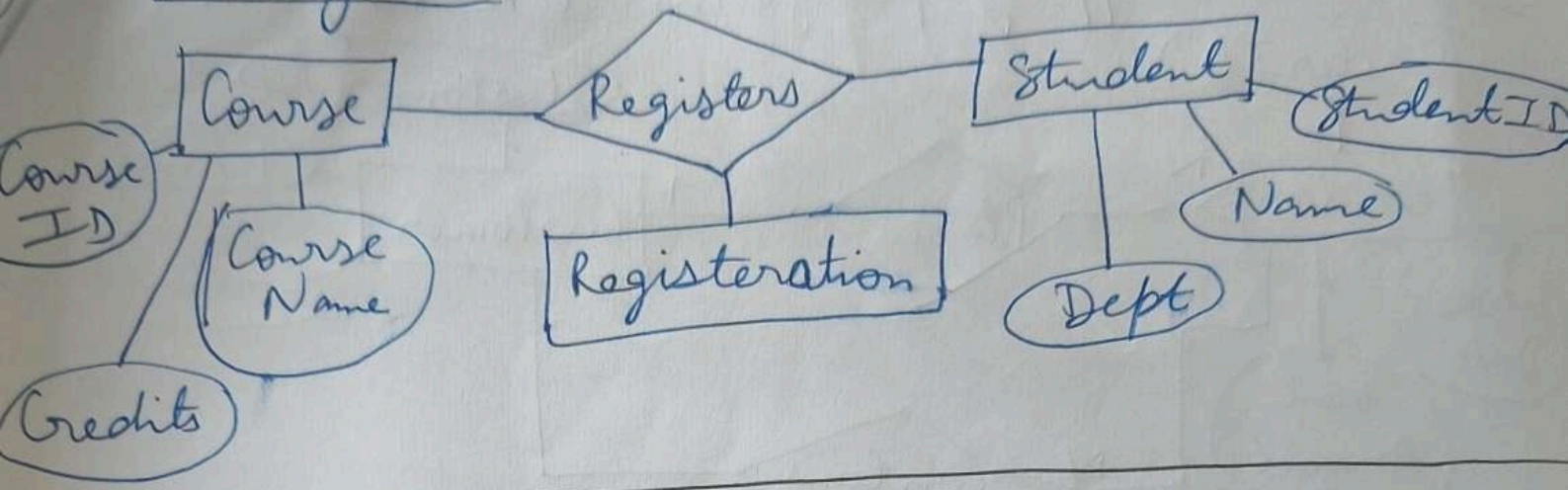


Q3. ER diagram for student - course Registration System.

Ans. 1) Entities → 1) Student (Student ID, Name, Dept).
2) Course (Course ID, Course Name, Credits).
3) Registration (Reg Date).

2) Relationship → Student Registers course (Many to many).

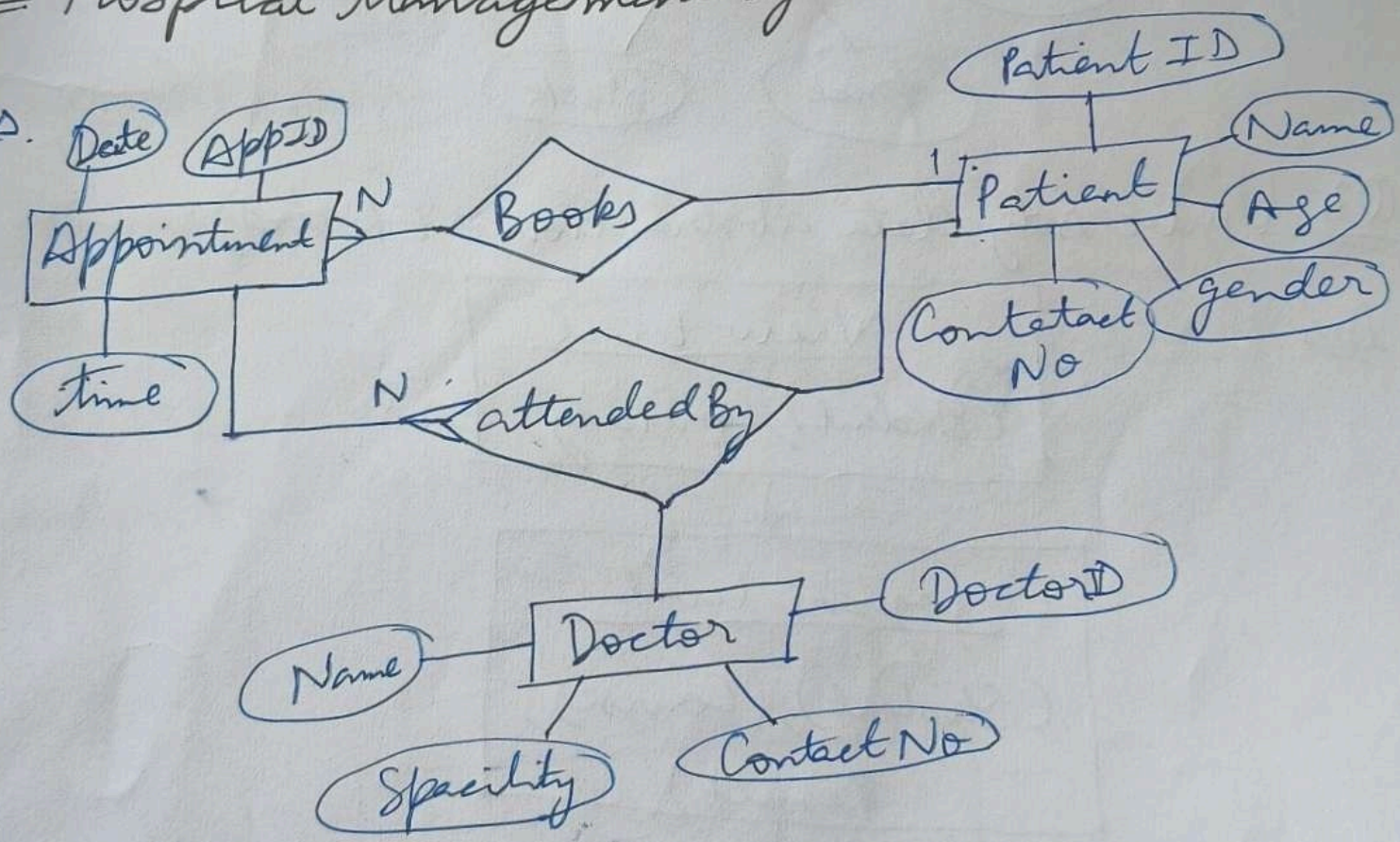
→ Diagram :-



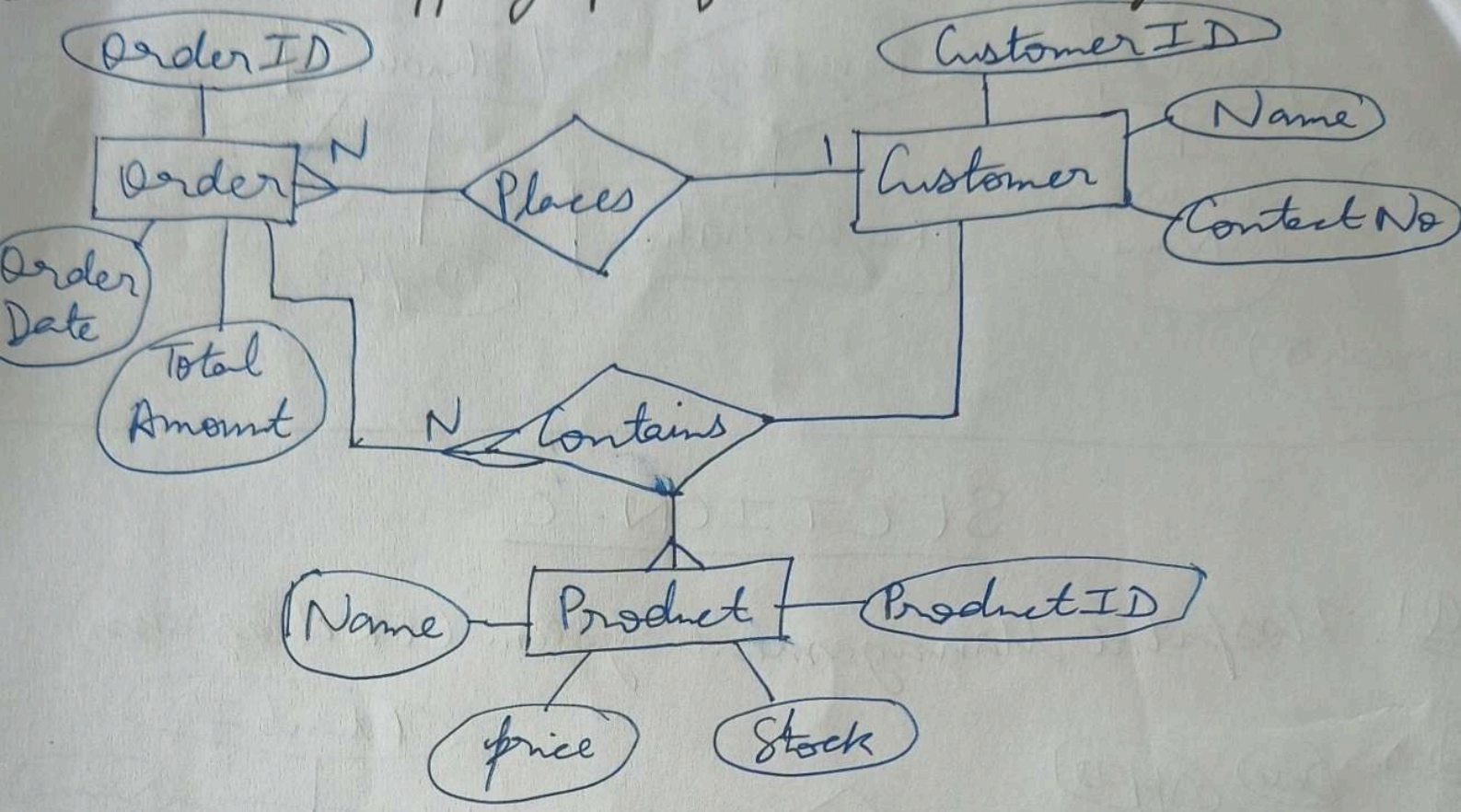
SECTION - C

Q1 Hospital Management System (ER diagram).

Ans.

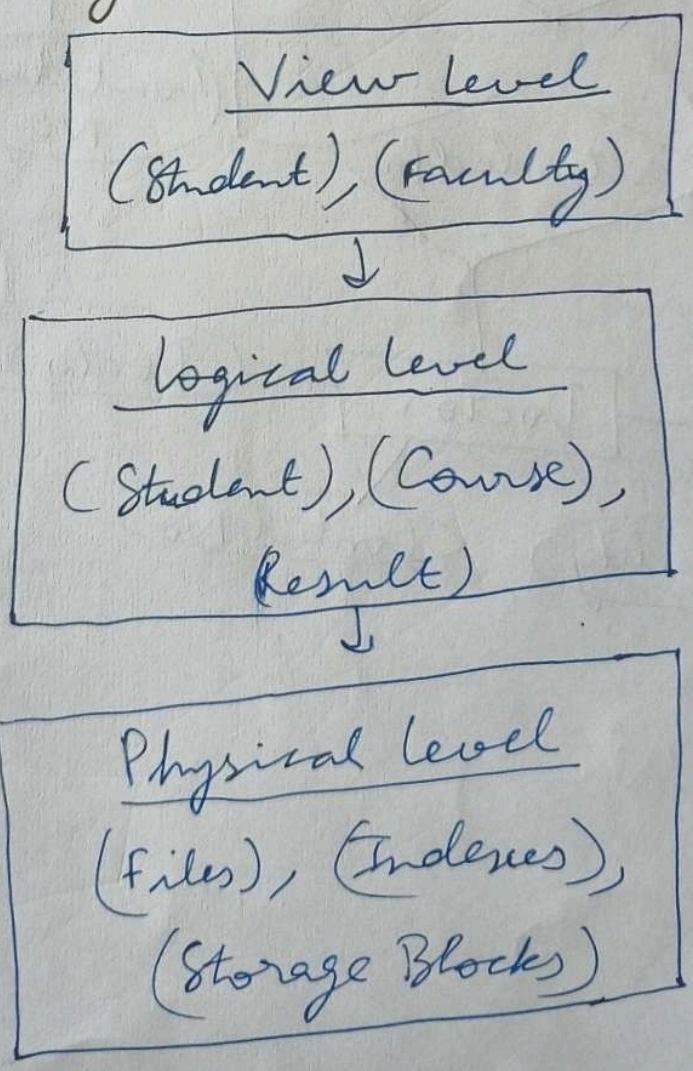


Q2. Online shopping platform – ER diagram.



Q3. University data abstraction – ER diagram.

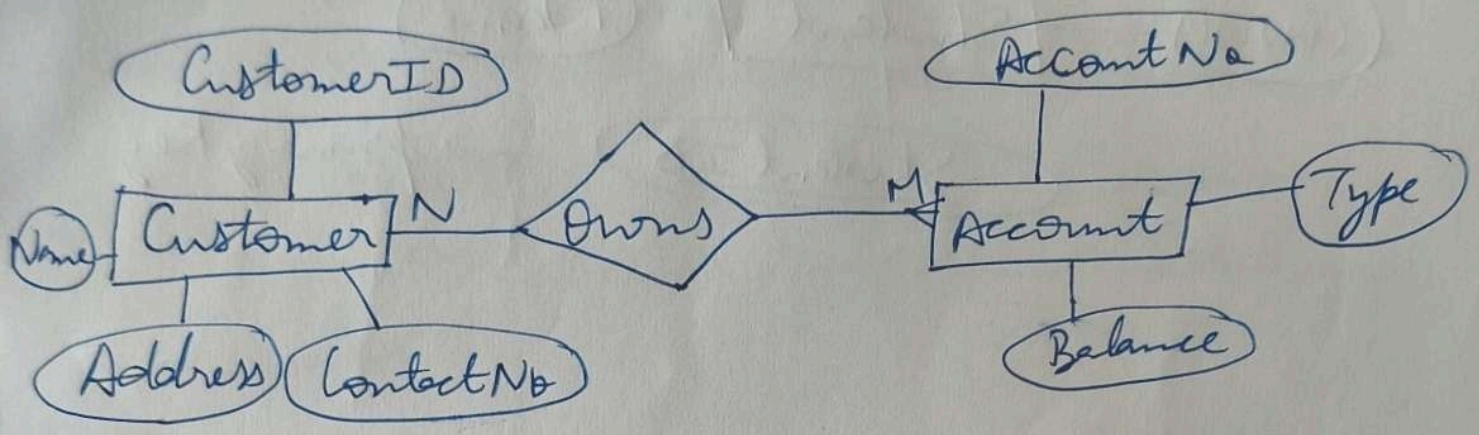
Ans.



SECTION-D

Q1. Draw a Simple ER diagram for any real life system & explain one key design decision.

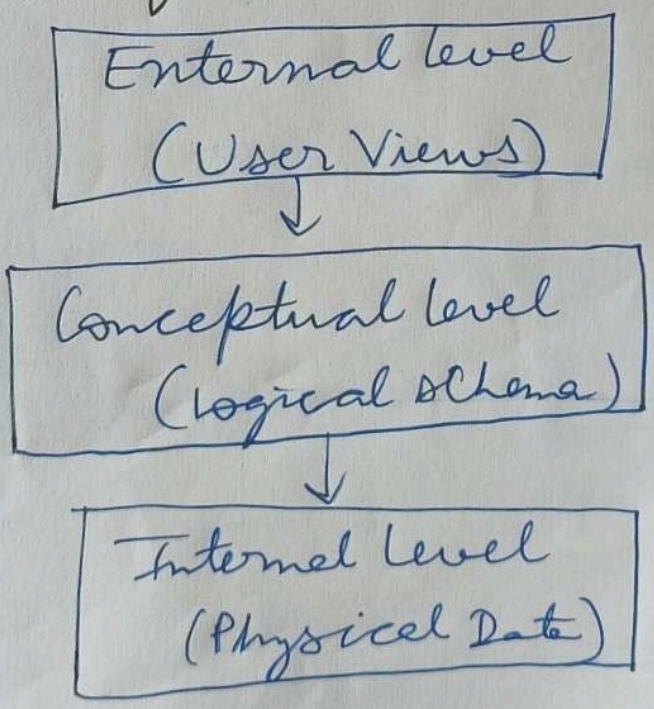
Ans Simple ER diagram of a Bank system.

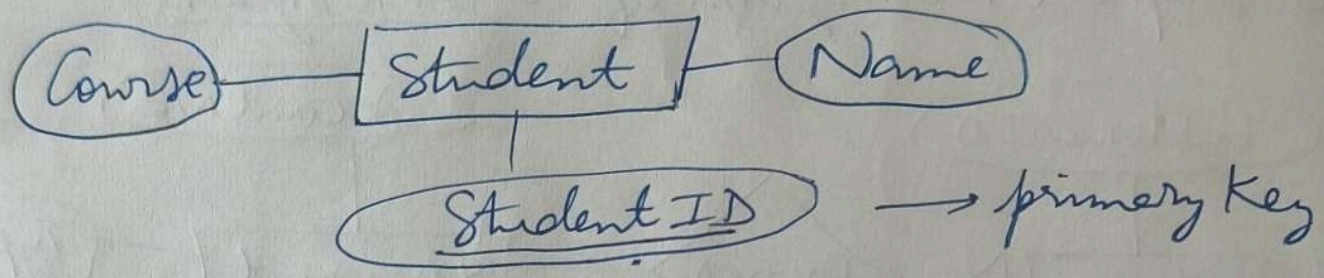


Here, Account is a seperate Entity to allow one customer to have multiple accounts.

Q2. Draw the three-schema architecture & mention one benefit.

Ans.



Q3. Draw an entity with attributes and indentify is primary key.

Ans. Student Entity with attributes :-